

Predicting Shelter Animal Outcomes at Intake Using Machine Learning

Date: 12th December 2025



Vinayak Mokashi

Github - [Link](#)



Introduction (Recap)

PROBLEM

- Overcrowding in shelters
- Importance – not enough adoptions because of overcrowding

TYPE OF PROBLEM

- Multi-class classification task
- Predict outcome type at intake (Only for Cats and Dogs)
- Low adoption probability – need extra support

DATA SOURCE AND COLLECTION

- Austin Animal Center website
- Outcomes and intakes data (CSV files) – merged
- Collected by staff during animal check-in/out
- 1st October 2013 – 2nd May 2025

DIFFICULTIES:

- Large dataset - 135k rows
- Missing values (Only in Categorical features)

EDA (Recap)

- Possible “color bias” in Adoption decisions
- Named Animals – Higher Adoption Rate
- Young animals are adopted far more easily than seniors
- Imbalance is strong in the target variable (Requires Stratified Splitting)

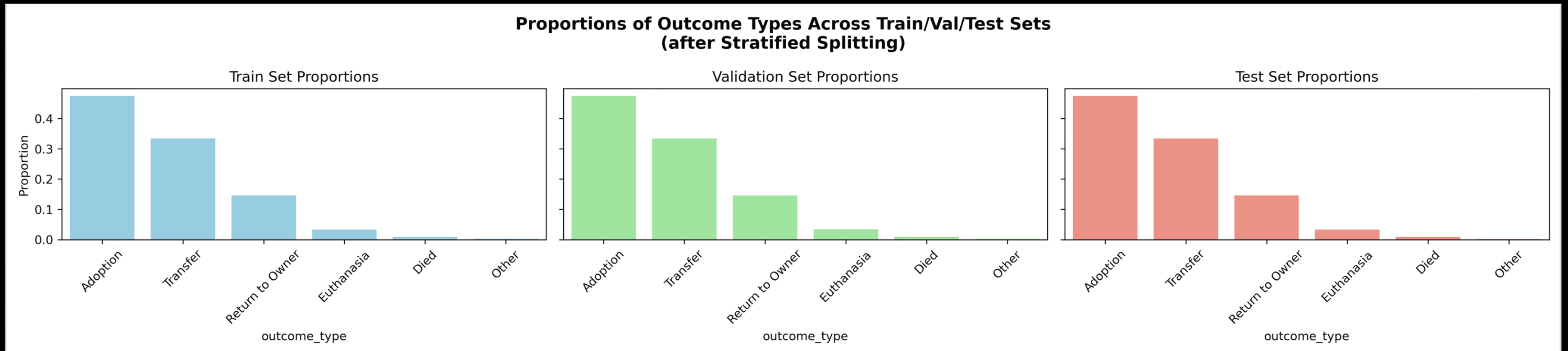
PREPROCESSING (Recap)

- Data sample: 10% random subsample of full data
- Continuous (2): scaled with StandardScaler
- Datetime (3): cyclical encoding (sin/cos)
- Categorical (8): one-hot encoded
- Before Preprocessing – 13, After Preprocessing – 950+

Cross Validation Pipeline

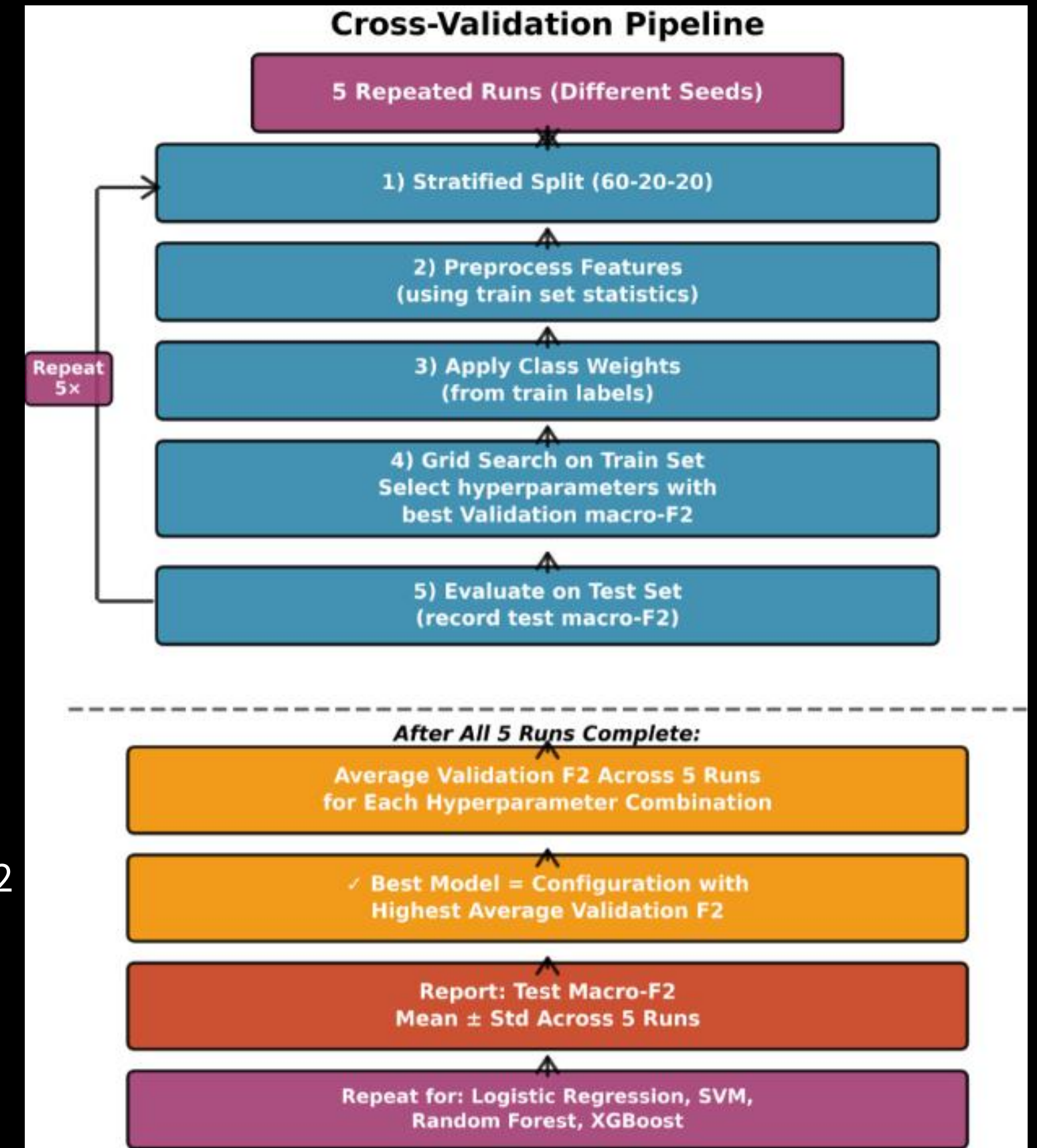
Splitting Strategy: Stratified Split

- Training – 60%
 - Validation – 20%
 - Test – 20%
- Data is i.i.d.
 - Stratified split preserves class distribution across splits (handles imbalance)
 - K-Fold is computationally expensive, single stratified split is more efficient



Cross Validation Pipeline

- 5 repeated runs with different random seeds
- For each run:
 - 1) Stratified split on outcome_type (60-20-20)
 - 2) Preprocess features using train set statistics
 - 3) Apply class weights (derived from train labels) to handle imbalance
 - 4) Grid search: train on Train set, select hyperparameters with best Validation Macro-F2
 - 5) Evaluate selected model on Test set
- Model selection for deployment:
 - ✓ Average validation Macro-F2 across 5 runs for each hyperparameter combination
 - ✓ Best model = configuration with highest average validation Macro-F2
- Performance reporting:
 - ✓ Test macro-F2: mean $\pm$ std across 5 runs
- Repeat for Logistic Regression, SVM, Random Forest, and XGBoost.



Cross Validation Pipeline

Models and Hyperparameters tried:

Model	Tuned Hyperparameters	Search Space
Logistic Regression (Elastic Net)	C [0.0001, 0.001, 0.01, 0.1, 1, 10, 100, 1000] l1_ratio [0.0, 0.2, 0.4, 0.6, 0.8, 1.0]	$8 \times 6 = 48$
SVM	C [0.001, 0.01, 0.1, 1, 10, 100, 1000] gamma [0.0001, 0.001, 0.01, 0.1, 1, 10, 100, 1000] kernel ['linear', 'rbf']	$7 + (7 \times 8) = 63$
Random Forest	max_depth [1, 2, 3, 5, 8, 10, 15, 20] max_features ['sqrt', 'log2', 0.1, 0.3, 0.5, 0.7, 0.9, 1.0]	$8 \times 8 = 64$
XGBoost	max_depth [1, 3, 5, 10, 20] reg_alpha [0.0, 0.01, 0.1, 1.0, 10.0, 100.0] reg_lambda [0.0, 0.01, 0.1, 1.0, 10.0, 100.0]	$6 \times 6 \times 5 = 180$

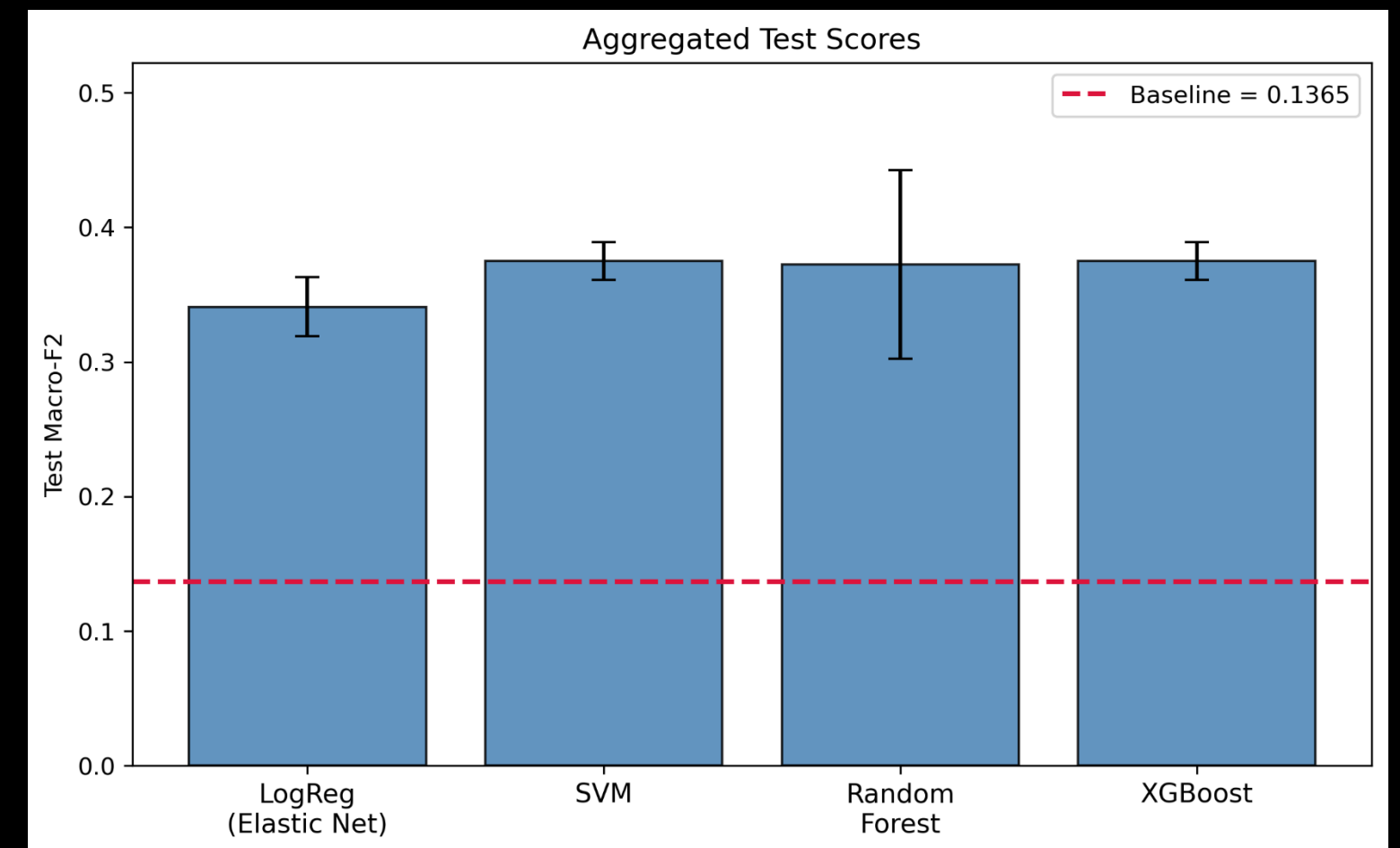
- Random Forest: n_estimators = 500
- XGBoost:
 - n_estimators = 10000 with early_stopping (50 rounds)
 - learning_rate = 0.03, colsample_bytree = 0.9, subsample = 0.66
- In each model, the class weights were adjusted (balanced)

Results

Evaluation Metric: Macro-F2 Score (Emphasizes Recall)

- Baseline score is calculated by predicting the majority class (Adoption) for all points
- Baseline Validation Macro-F2: 0.1365 ± 0.000
- Baseline Test Macro-F2: 0.1365 ± 0.000
- (No variation due to stratified split)

Model	Aggregated Test Score	Improvement from Baseline (+)
Logistic Regression (Elastic Net)	0.341 ± 0.022	0.2046
SVM	0.375 ± 0.014	0.2381
Random Forest	0.3723 ± 0.070	0.2358
XGBoost	0.375 ± 0.014	0.2381



Results

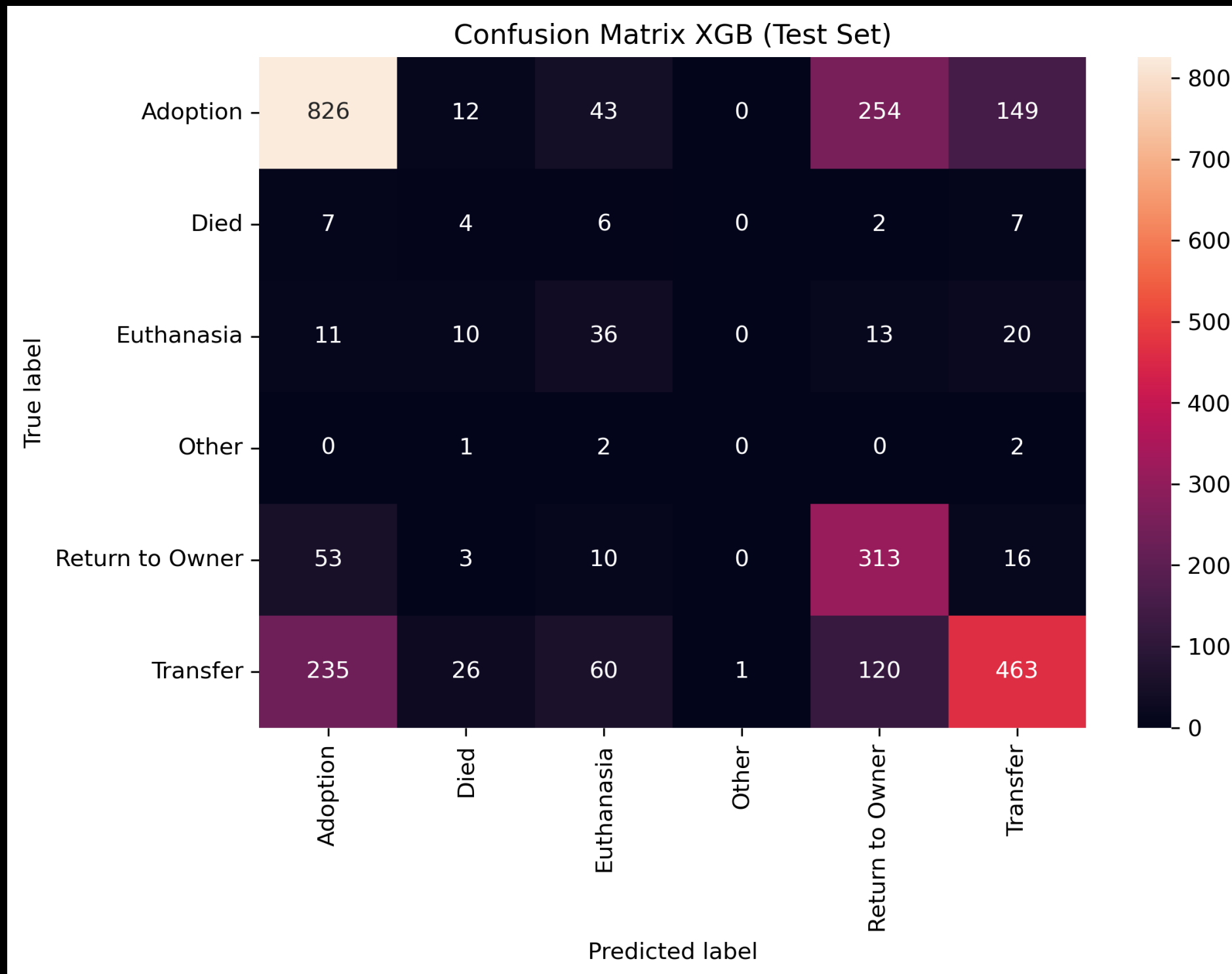
Best Hyperparameters for each algorithm:

Model	Best Hyperparameter	Best Average Validation Score	Improvement from Baseline (+)
Logistic Regression (Elastic Net)	C = 0.01 l1_ratio = 0.2	0.343 ± 0.010	0.207
SVM	C = 1 gamma = 0.1 kernel ['linear', 'rbf']	0.381 ± 0.018	0.245
Random Forest	max_depth = 20 max_features = 0.9	0.376 ± 0.006	0.239
XGBoost	max_depth = 3 reg_alpha = 0 reg_lambda = 0	0.387 ± 0.006	0.25

Model to deploy – XGBoost

- Mean Test Macro-F2 is ~17 standard deviations above the baseline Test Macro-F2
- Validation Macro-F2 is ~42 standard deviations above the baseline Validation Macro-F2

Results



- What steps did you take to make the model interpretable?
 - ✓ Analyzed the Confusion Matrix
 - ✓ Calculated Global Feature Importance using 3 ways:
 - 1) Permutation Feature Importance
 - 2) Total Gain
 - 3) Global SHAP
 - ✓ Local SHAP
- Strong on Return to Owner (high recall) and Adoption/Transfer (good recall)
- Severe weaknesses for Euthanasia, Died, Other (low/zero recall, many false negatives)
- Other: never predicted (model collapses for this class)
- Not deployment-ready for risk flagging
only tentatively useful for high-level capacity planning

Outlook

- Instead of using sampled 10% data, using the whole dataset should boost predictive power and reduce variance.
- Increase class weight / sampling for Euthanasia, Died, Other. Can also consider using SMOTE.
- Reframe the task to predict length of stay (or survival / time-to-outcome) and allocate resources for animals that are likely to stay long.
- Richer Feature Engineering for Accuracy + Interpretability:
 - Group color and breed features into few behaviorally / visually meaningful clusters.
 - Add context features such as shelter population size, population by animal type and age group
- Bias & fairness: Audit predictions by breed, color, age, and intake type to detect and reduce discrimination.



Thank you

ADOPT IF YOU CAN!